

Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [IT_2123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain how XOR gate can be realized using only NAND gates with neat diagram. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Write a Verilog code snippet for a 2-input OR gate using dataflow modeling. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Construct a circuit using only NAND gates for the function $F = AB + \bar{C}$. (2)
Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Design a Half Adder and Full Adder using logic gates and verify their operation. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Implement a 4-to-1 Multiplexer using logic gates and derive the Boolean expression. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
C) Illustrate how an encoder can be used to reduce the number of data lines in a system. (2)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply Verilog constructs to model a D flip-flop with synchronous reset. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Design a 4-bit Johnson Counter and discuss its applications. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Differentiate between asynchronous counter and synchronous counter in terms of timing. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Explain the Booth's multiplication algorithm with example and flow diagram. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Apply the concept of pipeline organization to show the role of pipeline registers in improving system performance. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Explain restoring and non-restoring division with an example. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Discuss the role of SISD in parallel architectures and give real-time examples. (5)

- A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Discuss the necessity of implementing pipelined architecture over non-pipelined architecture. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Construct the steps for floating-point addition according to IEEE 754 standard. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		IT_2123:DIGITAL SYSTEMS AND COMPUTER ORGANIZATION	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		IT_2123:DIGITAL SYSTEMS AND COMPUTER ORGANIZATION	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	24(48%)
4	Analysis	8	26(52%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)